

Metastasis Prediction in Colorectal Cancer with Whole Slide Images using Graph Attention Networks

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Abstract— We present an N0/N1 classification system for lymph node metastasis prediction in pT1 colorectal cancer from Whole Slide Images. Tissue is represented as a spatial graph over patch coordinates, preserving the real spatial topology of the tissue. Each node is a 2048×2048 px patch represented by a CLS embedding of shape $[1, 1536]$ from the UNI2 model. Two configurable architectures are compared: a GAT baseline with flat readout (mean, max, mean_max or attention) and a GAT+DiffPool model with hierarchical DiffPool interleaved between GAT layers for multi-scale representations. Patient-level diagnosis is obtained via MIL aggregation over histological sections. Hyperparameter search combines two Optuna studies (5-fold CV + 90/10 retraining of the best configurations). The best model is the GAT baseline, achieving test AUC = 0.875, sensitivity 100% and specificity 57.5%.

Keywords— pT1 colorectal cancer, Lymph node metastasis, Digital histopathology, Graph Attention Networks, Hierarchical DiffPool, Spatial Delaunay graph, Multiple Instance Learning.



1 INTRODUCTION

C OLORECTAL cancer (CRC) is one of the most serious public health challenges in our setting. In Spain, the SEOM projects around 44,132 new diagnoses for 2026, making it the most frequent tumor and the second leading cause of cancer death, with 14,629 deaths [1].

pT1 carcinoma is particularly complex: the tumor has invaded the submucosa but not the muscularis propria, and the risk of lymph node metastasis (LNM) ranges between 2% and 23% depending on the depth of invasion [2]. When poor-prognosis criteria are present (submucosal invasion $\geq 1000 \mu\text{m}$, vascular or lymphatic invasion, undifferentiated histology, *tumor budding* of grade ≥ 2 , or a positive vertical margin [3, 4]), the risk rises to around 10%, leading clinical guidelines (JSCCR [3], NCCN, and ESMO) to recommend additional radical surgery. Since most pT1 tumors do not develop real LNM, it is estimated that up to 80% of these surgeries are unnecessary [5, 6].

Digital histopathology emerged in the late 1990s with the first whole-slide scanners, and since the mid-2010s deep learning models have been applied to it to assist the pathologist [7]. This work, developed at the CVC [8] under the

supervision of Dr. Debora Gil, addresses the N0/N1 classification task from *Whole Slide Images* (WSI) by modeling each tissue section as a spatial graph. A GAT baseline with flat *readout* is compared against a model with hierarchical *pooling* (DiffPool) interleaved between layers, and different MIL (*Multiple Instance Learning*) aggregation strategies are proposed for patient-level diagnosis.

Two reasons motivate this work. From a clinical standpoint, AI systems can consistently detect subtle patterns (often outperforming clinical guidelines) and help reduce unnecessary surgeries. From a technical standpoint, combining graph attention networks with deep learning on real-world data sits at the intersection of graph theory and deep learning that I want to explore within Data Engineering.

2 STATE OF THE ART

2.1 Clinical Guidelines and Predictive Models for pT1 CRC

The reference clinical guidelines (JSCCR [3], NCCN, and ESMO) recommend additional surgery when the patient presents one or more of the histopathological risk criteria mentioned in the Introduction. Despite their high sensitivity, their specificity is low: in a comparative study of 560 patients [9], the reported AUCs are 0.596 (JSCCR), 0.749 (NCCN), and 0.743 (ESMO)¹, with specificities of

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¹Since these are binary classifiers, a guideline's AUC equals $(\text{sens} + \text{spec})/2$ (*balanced accuracy*).

19% (JSSCR), 52% (NCCN), and 50% (ESMO): following JSSCR, more than 80% of operated patients did not have real LNM [9, 5].

In parallel, Piao et al. [10] proposed an AI model over clinicopathological variables ($n = 546$ training, $n = 105$ test) based on gradient boosting (*LightGBM*), achieving 100% sensitivity, 85.8% specificity, an AUC of 0.960, and 92.9% *balanced accuracy*; the rate of patients classified as high risk dropped from 83.8% (JSSCR) to 20.9%. Dawson et al. [4] proposed the IBC Score, a weighted score over lymphovascular invasion, *tumor budding* grade, invasion depth, and tumor grade, achieving an AUC of 0.68 over 565 patients. Unlike these works, this thesis performs the prediction exclusively from the histopathological image (WSI), without structured clinical variables or manual pathologist scoring.

2.2 WSI Classification with MIL

Whole Slide Images have resolutions of up to $150,000 \times 150,000$ pixels and cannot be processed directly with conventional networks. The usual approach is to divide the slide into small *patches* and work with MIL, where the model must infer the global label (N0/N1) from fragments without individual labels. Ilse et al. [11] proposed attention-based aggregation (*attention-based MIL*), which learns which *patches* are most relevant for diagnosis. Lu et al. [12] applied this method (CLAM) to lymph node metastasis detection in the CAMELYON challenge (breast cancer, AUC ~ 0.95); that task, however, detects metastasis *present* in the lymph node's own slide, whereas here the nodal status is predicted from the primary tumor, so the results are not directly comparable. Li et al. [13] proposed DSMIL, which combines two branches, at the *patch* and slide level, with contrastive learning. The prediction of lymph node metastasis from histopathology has been studied in other cancers: Muti et al. [14] show that it is possible to predict N0/N1 status from gastric cancer slides, with promising results.

2.3 Graphs and GNNs in Digital Pathology

Graph neural networks (GNN) make it possible to treat tissue as a structure where regions are connected according to their proximity, capturing spatial relations that convolutional neural networks (CNN) do not directly model. Jaume et al. [15] presented HistoCartography, a framework for graph analysis in digital pathology, with results showing improvements over CNNs in biopsy classification. This framework distinguishes between *cell-graphs* (nodes = individual cells) and *tissue-graphs* (nodes = tissue regions); this work is closer to the second type, with nodes representing tissue *patches* rather than cells. Regarding architecture, HistoCartography classifies these graphs with GIN (*Graph Isomorphism Networks*) networks, whereas this work opts for GAT [16], which add an attention mechanism that lets the model weight each neighbor differently, particularly useful in histopathology where not all tissue regions carry the same information. Regarding graph *pooling*, Grattarola et al. [17] review the main existing methods, including DiffPool [18], which groups nodes into super-nodes progressively.

A key decision in building these graphs is the edge criterion: some works base them on *feature similarity* between *patches*, while others opt for spatial proximity between tissue regions. This work follows the second approach and connects *patches* via a triangulation over the tissue coordinates, so that nodes and edges preserve the real topology rather than purely semantic relations.

2.4 Foundation Models for Histopathology

In recent years, *foundation models* pretrained specifically on pathology data have emerged, generating embeddings far more informative than generic CNNs. UNI2 [19] is a *vision transformer* (ViT) trained on more than 100 million *patches* from 20 tissue types, with good results across numerous histopathological classification tasks. CONCH [20] is a similar model that additionally incorporates image-text alignment. This project uses UNI2 embeddings as the feature vector for each node, since they are pretrained in the same domain and capture relevant morphological information.

2.5 Prior Work at the CVC on pT1 CRC

Martí [21] compared CNNs, ViTs, and GNNs on the same pT1 CRC dataset to classify individual *patches* by percentage of tumor infiltration, obtaining the best AUC (0.954 ± 0.077) with a hybrid ViT+NN architecture. That task, however, is tissue classification at the *patch* level, without spatial topology or patient-level diagnosis, and does not predict nodal status; this thesis builds on UNI2 CLS embeddings to construct spatial graphs of the sections and predict N0/N1 status from a patient-level MIL perspective.

3 OBJECTIVES

The general objective of this work is to **classify N0/N1 nodal status** in pT1 colorectal cancer from real WSIs, using the UNI2 model's CLS embeddings and representing each tissue section as a spatial graph (Delaunay triangulation over the tissue coordinates).

From this general objective the following specific objectives are derived:

- Build a complete *pipeline*, from generating spatial graphs from CLS embeddings to training and evaluating the model.
- Compare a GAT baseline with flat *readout* against a model with hierarchical DiffPool interleaved between GAT layers, assessing whether progressive graph reduction improves intra-section representation.
- Implement and compare patient-level MIL aggregation strategies to integrate predictions from multiple histological sections.

4 DATA AND PREPROCESSING

This project is carried out in collaboration with a UAB team that covers everything from data preprocessing to the creation of alternative autoencoders for feature extraction, for subsequent classification using various models.

4.1 Input Data

WSI images. The original data consist of whole-slide images (WSI) obtained with high-resolution scanners and stained with hematoxylin-eosin, stored in MIRAX format (.mrxs). Dimensions range between 30,000 and 150,000 pixels per side, with a significant amount of empty space or artifacts that require prior treatment.

The effective dataset comprises 1,607 graphs (one per histological section), from 367 patients distributed unevenly between the two classes (Table 1). Of a total of 606 patients in the clinical database, 236 are discarded due to indeterminate diagnosis (NX), and of the remaining 370 with a valid diagnosis, 3 lack generated UNI2 embeddings, leaving 367 usable patients. Samples are split into a training+validation pool (85%) and a fixed test set (15%), both stratified at the patient level. Within the pool, 5-fold cross-validation is applied for hyperparameter search (Section 6.1).

Annotations and masks. Dr. Eva Musulén, a specialist pathologist, manually performed the annotations using *Labelme* software, identifying three categories of involvement: *infiltration* (abnormal tissue invading regions it should not), *dysplasia* (abnormal cells that may become cancerous), and a *combination* of both (used in cases of diagnostic doubt). From these, color-coded segmentation masks are obtained: red (infiltration), blue (dysplasia), and light green (combined); Figure 7 (Appendix A.1) shows an example.

4.2 Graph Construction

From patch to node. A non-overlapping sliding window is applied to divide each slide into 2048×2048 pixel *patches* at full resolution. For each window, the percentage of tissue in the corresponding region of the annotated binary mask is checked; windows with no tissue pixels are discarded. The UNI2 model [19] processes each *patch* individually and extracts the *CLS* token from its output, obtaining a vector of shape $[1, 1536]$ that condenses the patch’s morphological and textural information. Each *CLS* vector becomes a graph node, retaining the patch coordinates for edge construction.

The section graph. Formally, each histological section is represented as a graph $\mathcal{G} = (\mathbf{X}, \mathbf{A})$, where $\mathbf{X} \in \mathbb{R}^{N \times 1536}$ is the feature matrix of the N nodes (*patches*) and $\mathbf{A} \in \{0, 1\}^{N \times N}$ the adjacency matrix built via Delaunay triangulation followed by edge pruning. Each section generates an independent graph; a patient’s S sections are processed separately and the probabilities $\{P(N1_i)\}_{i=1}^S$ are aggregated via MIL (Section 5.5). As an alternative, a per-patient *mega-graph* was explored (block-diagonal over all sections, with no edges between them), but it gave worse results, likely because it forces a single global *pooling* over physically disconnected tissues.

Delaunay connectivity. Patch centers form a regular grid, but connecting them with a fixed grid is not enough: the mask leaves irregular tissue gaps and does not connect nearby regions at angles other than 90° . For this reason, nodes are connected via a **Delaunay triangulation** over their coordinates, which adapts to any distribution. Edges are then pruned in two steps: those that cross background pixels (mask with RGB value = 0) are removed, so as not to join regions separated by empty space, as are those that ex-

ceed twice the mean edge length, to avoid overly long connections. The result (Figure 1, with pruned edges in red) is a local mesh between neighboring *patches*.

5 MODEL ARCHITECTURE

5.1 Pipeline overview

At a high level, the system can be summarized in five chained blocks (Figure 2): **(1) Preprocessing**, which converts each WSI into a spatial graph (2048×2048 *patch* extraction, *CLS* embedding [1, 1536] with the UNI2 *foundation* model, and Delaunay graph construction with edge pruning); **(2) Architecture**, where GAT layers propagate information across the graph according to one of the two compared variants (flat *baseline* or with hierarchical DiffPool); **(3) Readout**, the final read-out that collapses the graph into a single embedding (mean, max, mean_max, or attention), identical for both architectures; **(4) MIL**, which aggregates the predictions of a patient’s sections (mean, max, noisy-OR, or attention); and **(5) Decision**, where the threshold t^* derived from the *validation set* is applied to obtain the binary N0/N1 classification.

The following sections detail each block: the GAT architecture and its two variants, the unified *readout*, MIL aggregation, and the training procedure.

5.2 GATClassifier

The implemented model, *GATClassifier*, is a graph classifier based on **Graph Attention Network** layers (*GAT-Conv* [16, 22]) for binary N0/N1 classification. The architecture is fully parametrizable: the number of GAT layers, the hidden dimension, and the number of attention heads are chosen as hyperparameters, and the whole model is trained *end-to-end* (a single backpropagation pass propagates gradients from the final MLP head down to the first GAT layers). Building on this common base, two architecture variants are compared, described in Section 5.3: a *baseline* and one with hierarchical DiffPool.

Attention mechanism. Unlike classical convolutional networks on graphs that treat all of a node’s neighbors uniformly, GAT learns to assign differentiated importance weights to each neighbor. Given the representation $\mathbf{h}_i$ of node i , for each pair of neighboring nodes (i, j) an attention coefficient is computed:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j])$$

where $\mathbf{W}$ is a learnable linear transformation, $\parallel$ is concatenation, and $\mathbf{a}$ a learnable vector. The coefficients are normalized with *softmax* over all neighbors:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}$$

and the new node representation is obtained as the weighted sum:

$$\mathbf{h}'_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j$$

Multi-head and non-linearity. To increase expressive capacity, each layer uses H independent attention heads,

TABLE 1: DATASET COMPOSITION (26 HOSPITALS). EACH GRAPH CORRESPONDS TO A HISTOLOGICAL SECTION.

Split	Patients			Graphs (sections)		
	N0	N1	Total	N0	N1	Total
Pool (train+val CV)	224	87	311	1133	219	1352
Test	40	16	56	212	43	255
Total	264	103	367	1345	262	1607

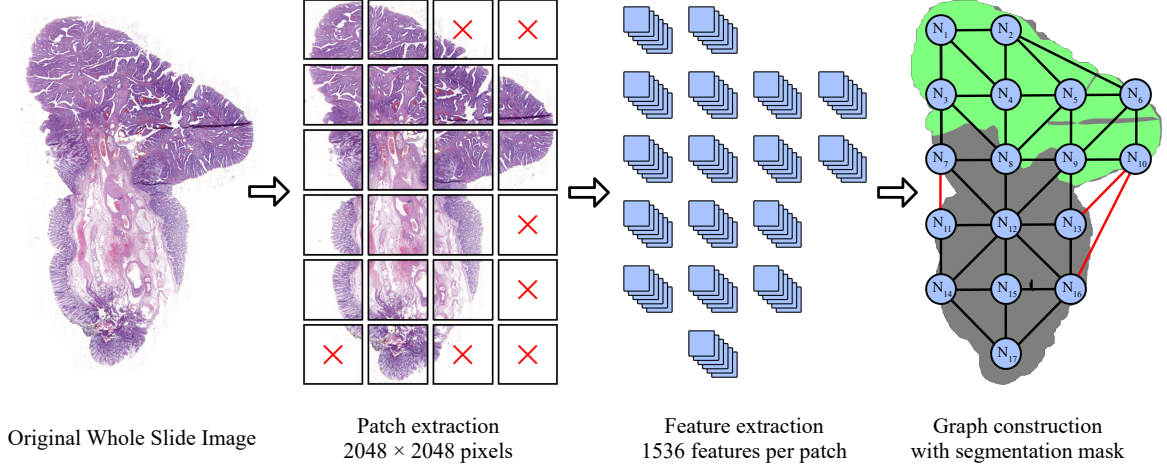


Fig. 1: Preprocessing stages for graph construction. From left to right: original WSI → 2048×2048 px *patch* extraction via sliding window + filtering → CLS embedding extraction of shape [1, 1536] via UNI2 (one vector per *patch*) → construction of the Delaunay graph over the annotated mask, where red edges have been removed by the mask filter

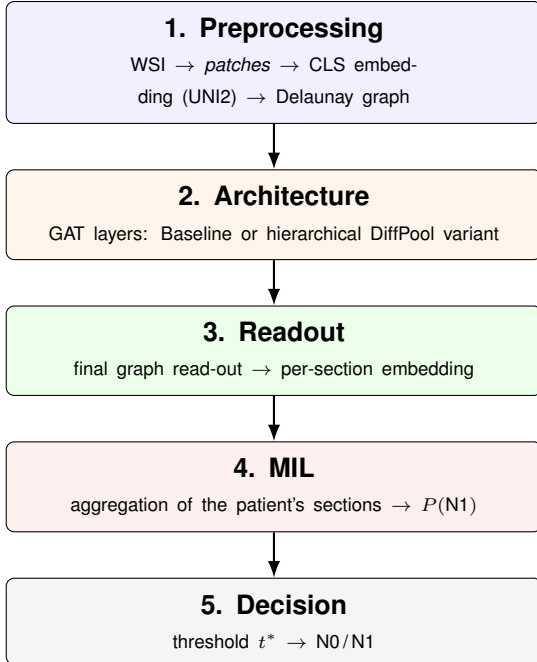


Fig. 2: Overview of the five-block *pipeline*.

parametrized by $\mathbf{W}^{(h)}$ and $\mathbf{a}^{(h)}$. The outputs are concatenated at intermediate layers and averaged at the last one. The *GATConv* layer applies no final activation within the layer itself; the non-linearity is applied externally as *BatchNorm* → ELU → *Dropout*. The only *internal* non-linearity is the *LeakyReLU* in the computation of e_{ij} .

5.3 The two architectures: Baseline and Diff-Pool

Building on the GAT layers described above, two architectures are compared that differ in how they process the graph before the final *readout*. Figure 3 shows one instance of each, and Appendix A.3 presents a detailed diagram with all configurable axes.

GAT Baseline (flat). Stacks L consecutive GAT layers (L tunable) that always operate on the graph's original N nodes, without reducing it. All spatial information is preserved up to the final *readout*.

GAT+DiffPool (hierarchical). Interleaves **DiffPool** [18] blocks between GAT layers. The idea is to group nodes into *super-nodes* and replace the graph with a smaller one; repeating this several times yields a multi-scale representation of the tissue. Each block learns, for each node, a *soft assignment* to the K super-nodes of the next level: instead of sending each node to a single group, it distributes membership across all groups with weights summing to 1 (the output of a small per-node MLP followed by *softmax*). From this assignment, node *features* are aggregated as a weighted average to form those of the super-nodes, and the original edges are projected onto the new graph, so the graph goes from N to $K \ll N$ nodes while retaining a compressed version of its structure. The number of interleaved blocks is an optimization axis, L_{DP} , with one GAT layer before, between, and after each one (total $L_{DP} + 1$ GAT layers).

DiffPool auxiliary loss. For each block, two regularization terms are added over the learned assignment: one that pushes neighboring nodes to be grouped together, and another (entropy-based) that makes each node assign cleanly to few groups. The total loss is the classification *cross-*

entropy plus λ_{aux} times these terms, with λ_{aux} as an optimization axis.

As an alternative to DiffPool, SAGPool [23] selects the K most relevant nodes via self-attention scores (discrete selection, not soft assignment); DiffPool was preferred for its continuous expressiveness and the auxiliary regularization loss.

5.4 Readout: final graph read-out

Whatever the architecture, the last step before classification is the same: a global *readout* that collapses the final set of nodes into a single vector $\mathbf{h}_s$ representing the whole section. The only difference is which nodes it is applied to: in the Baseline these are the N nodes of the last GAT layer, and in DiffPool the final K_{DP} super-nodes. The *readout* operation is thus an axis shared by both branches, with four configurable options: *mean*, *max*, or *mean_max* (fixed aggregations), and *attention*, which learns a scalar weight per node via a linear layer followed by *softmax* and aggregates via weighted sum.

Finally, the MLP head (*Linear* $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear) transforms this global representation $\mathbf{h}_s$ into the section's N0/N1 logits.

5.5 Patient-Level Aggregation (MIL)

Diagnosis is per **patient**, not per slide. A given histological section may show no visible signs of metastasis when analyzed in isolation, while another section from the same patient does show it. If trained at the slide level, N1 sections from N1 patients that show no visible metastasis would receive an incorrect supervision signal (N0), contaminating training.

To avoid this problem, the **Multiple Instance Learning** paradigm is used (MIL [24]): each patient is treated as a *bag* and all of their slides share the same label. The model learns to classify the patient from their sections, without needing individual per-slide labels.

In all cases, the MLP head has the same structure: *Linear*(h) $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear(2), where h is the hidden dimension (hyperparameter) and the output is a pair of logits (N0/N1); the weights are learned via backpropagation. The difference between strategies is *when* it is applied:

- **Mean, Max, Noisy-OR:** the MLP is applied **per section**, producing $P(\text{N1})_s$ for each of the patient's S sections; the probabilities are aggregated as scalars:

- **Mean:** $P(\text{N1}_p) = \frac{1}{S} \sum_i p_i$.
- **Max:** $P(\text{N1}_p) = \max_i p_i$.
- **Noisy-OR:** $P(\text{N1}_p) = 1 - \prod_i (1 - p_i)$.

- **Attention MIL** (*gated attention*, Ilse et al. [11]): the embeddings of the S sections ($\mathbb{R}^h$ each) are first aggregated into a single patient vector via learned attention weights (weighted sum of the embeddings); the MLP is applied **only once** to this vector to obtain the N0/N1 logits. The internal dimension of the attention module is fixed (128) and is not part of the hyperparameter search.

6 EXPERIMENTAL DESIGN

6.1 Data and evaluation scheme

The dataset is split into an 85% pool and a 15% test set, stratified at the patient level (Figure 4). All search (hyperparameter and threshold selection) is done only on the pool, so the test set plays no part in any decision and provides an unbiased performance estimate. The procedure has three stages:

1. **Test (15%, 56 patients):** separated from the moment the graphs are built and held out for evaluating the best models per architecture.
2. **Bayesian search + 5-fold CV over the pool (85%, 311 patients):** each configuration is evaluated with per-patient *Stratified K-Fold* and *early stopping* (*patience* = 20), independently for each architecture. The search uses the *Tree-structured Parzen Estimator* (TPE) over ~ 15 axes: the first 200 *trials* ($\sim 10 \times$ dimensions) are random uniform sampling (exploration), and afterward TPE exploits the history. The maximized objective is $\overline{\text{spec}}_{\text{CV}} - 10 \cdot \max(0, 1 - \overline{\text{sens}}_{\text{CV}})$: mean specificity, strongly penalizing drops in sensitivity (Appendix A.6).
3. **Final retraining and test evaluation:** with the best hyperparameters for each architecture, the pool is split again 90/10 (internal train/val for *early stopping*), and the final model is evaluated **once** on the test set (15%) with the threshold t^* , the median of the 5 t^* values obtained across the CV *folds*.

6.2 Class Imbalance and Operating Threshold

The dataset shows a strong imbalance (approximately 72% N0 and 28% N1). In metastasis detection, a false negative is more serious than a false positive: the clinical goal is to maximize specificity while keeping sensitivity $\approx 100\%$, to reduce unnecessary surgeries without missing any N1 case. Three complementary mechanisms are applied:

- **Stratified split** by patient (keeps the N0/N1 ratio in each subset).
- **Cross-entropy weighting** with a positive weight w_+ (gradient multiplier for the N1 class) as an optimization axis, $w_+ \in [1.5, 8]$.
- **Operating threshold derived from val (t^*):** two criteria were compared: (T1) fixed threshold $t = 0.5$; and (T2) the maximum threshold on the val ROC curve that keeps TPR = 1 (zero false negatives on validation). For T2, the threshold is computed on each *fold* and its median is applied to the test set.

6.3 Evaluation Metrics

The main metrics are AUC-ROC (overall ranking, threshold-independent) and the clinical metrics at the operating threshold: sensitivity ($\text{TP}/(\text{TP} + \text{FN})$), specificity ($\text{TN}/(\text{TN} + \text{FP})$), positive predictive value (PPV, $\text{TP}/(\text{TP} + \text{FP})$), and negative predictive value (NPV, $\text{TN}/(\text{TN} + \text{FN})$),

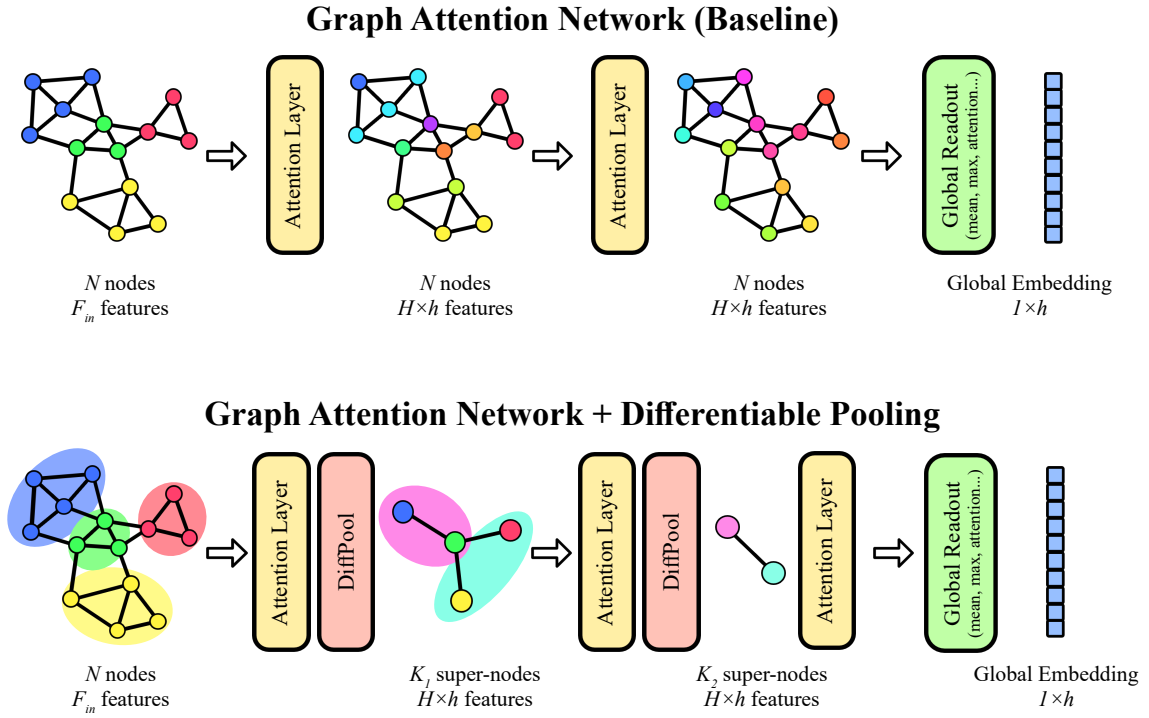


Fig. 3: Visual comparison of the two configurable architectures (one instance of each branch with two processing layers). **Top:** GAT Baseline, where two attention layers operate on the N original nodes and a global *readout* produces the final embedding. **Bottom:** GAT+DiffPool with two DiffPool blocks ($L_{DP} = 2$, i.e. three interleaved GAT layers), which progressively collapse the graph down to, in this case, two final super-nodes.

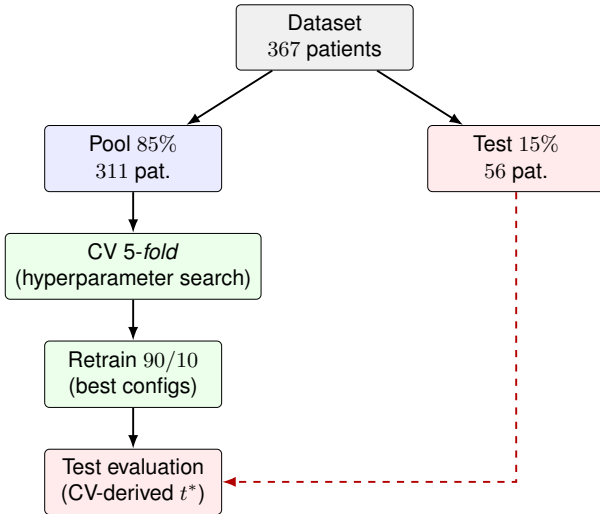


Fig. 4: Split and evaluation scheme. The test set (15%) is isolated from the start and used only at the end (dashed arrow).

where TP, TN, FP, and FN are true/false positives and negatives.

Clinical interpretation. If the model’s output were used to decide on surgery (operating on cases predicted as N1), sensitivity is the percentage of N1 patients who would receive treatment (a sensitivity of 100% means no patient with metastasis would go untreated), and specificity is 1 minus the percentage of N0 patients operated on unnecessarily.

6.4 Tools and Reproducibility

The implementation follows a modular design: graph construction, the model, the training code, and the Bayesian *sweep* are decoupled, and configuration is managed via YAML files. The search is orchestrated with *Optuna*, with persistent, resumable *SQLite* storage (no *trial* is lost if the process is interrupted). To ensure reproducibility, all *sampler* draws, *CV folds*, and weight initialization use a fixed *seed*. The code is available in the project’s public repository [25], which includes as a submodule the model and search code, developed jointly with other team members [26].

7 RESULTS

Two isolated per-architecture Optuna TPE studies were optimized, and the best configurations of each branch (GAT Baseline and GAT+DiffPool) were re-evaluated following the procedure in Section 6.1 (5-fold CV + 90/10 retrain + test evaluation). Table 2 summarizes the most relevant ones.

Reading the results. **GAT Baseline clearly outperforms GAT+DiffPool** on this dataset: the best DiffPool reaches a test AUC of 0.786, 9 points below the Baseline (0.875). Sections have only a few hundred *patches*, and hierarchical reduction to super-nodes discards more information than it synthesizes. Moreover, only the Baseline maintains $\text{sens.} = 100\%$ on the test set with the t^* threshold derived from val.

Selected model. The final model is a GAT Baseline with *attention readout*, *mean MIL*, $h = 256$, $H = 4$, *dropout* 0.3, $\lambda_{wd} = 10^{-4}$, $\eta = 3 \cdot 10^{-4}$, and Adam with *ReduceLROnPlateau*. It achieves AUC CV 0.862 ± 0.038 , test AUC

TABLE 2: BEST CONFIGURATIONS PER ARCHITECTURE, COMPARED BY AUC UNDER 5-fold CROSS-VALIDATION.

Architecture	Readout	MIL	h	H	dropout	AUC CV
Baseline	attention	mean	256	4	0.3	0.862 ± 0.038
DiffPool	mean_max	mean	128	2	0.44	0.847 ± 0.048

0.875, sensitivity 100% (16/16 N1), and specificity 57.5% at threshold $t^* = 0.133$.

7.1 Hyperparameter Importance

The *f*ANOVA importance analysis for each study (full table in Appendix A.4) shows that MIL aggregation is the most determinant axis of performance for the Baseline ($I = 0.64$; no other axis exceeds 10%), while for DiffPool the importance is spread between MIL (0.22), the auxiliary weight λ_{aux} (0.14), and the *learning rate* (0.13).

7.2 Comparison with Clinical Guidelines and SOTA

Table 3 positions the selected model against the reference clinical guidelines (Section 2.1) and two AI models (one over clinical variables and another over WSI images). The guideline figures come from [9] ($n = 560$), while the GAT figures were obtained on this work’s test set ($n = 56$; N1 prevalence 28.6%, higher than the real clinical rate of ~ 8 –15% [5]). Sensitivity and specificity are independent of prevalence and form the basis of the comparison; PPV must be interpreted with caution (Section 8.2).

Since a false negative (failing to treat an N1 case) is clinically unacceptable, **T2 is the only acceptable option**: T1 ($t = 0.5$) leaves 7 of 16 N1 patients undetected ($\text{FN} = 7$), while T2 ($t^* = 0.133$, $\text{TPR} = 1$) guarantees zero false negatives ($\text{sens.} = 100\%$, $\text{NPV} = 1.000$). The resulting specificity (57.5%) exceeds the JSCCR guideline (19%) and the IBC Score (26%), and is comparable to NCCN (52%) and ESMO (50%). Figure 5 shows the ROC curves of the CV’s 5 *val sets* and the individual operating points; the median of the 5 thresholds is $t^* = 0.133$. Figure 6 overlays the test ROC curves with the clinical guidelines.

Clinical impact. The t^* threshold maintains $\text{sens.} = 100\%$ on the test set (16/16 N1 detected, $\text{NPV} = 1.000$), with $\text{spec.} = 57.5\%$. The literature estimates that 80–90% of current pT1 surgeries are unnecessary [6], consistent with the 19–52% specificities of the guidelines. While maintaining $\text{sens.} = 100\%$, the GAT would unnecessarily operate on $\sim 42.5\%$ of N0 patients, versus $\sim 81\%$ (JSCCR) or $\sim 49\%$ (NCCN/ESMO). This represents a $\sim 47\%$ **reduction in unnecessary surgeries** relative to JSCCR (and a $\sim 13\%$ reduction relative to NCCN/ESMO), **without leaving any N1 patient untreated** ($\text{sens.} = 100\%$). At threshold 0.5, sensitivity is lower (56.2%); t^* is what makes the model clinically useful.

8 DISCUSSION AND CONCLUSIONS

8.1 Discussion

The model’s test AUC (0.875) exceeds that of JSCCR (0.596), NCCN (0.749), and ESMO (0.743) [9]. At thresh-

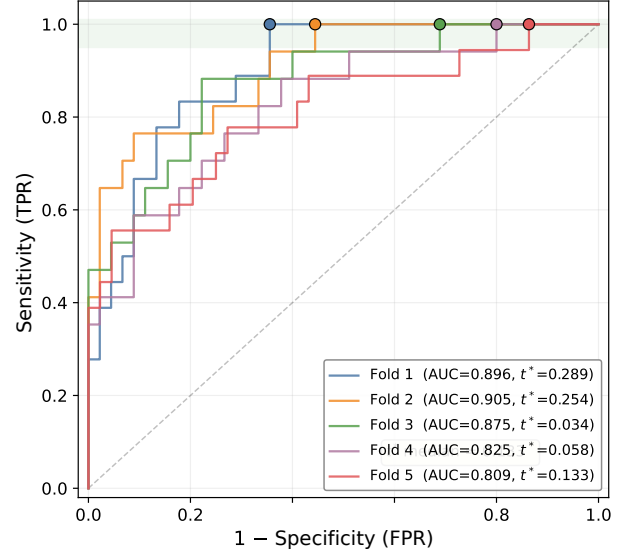


Fig. 5: ROC curves of the 5 cross-validation *folds* (GAT Baseline). Each circle marks the maximum $\text{TPR} = 1$ threshold of its *fold*; the median of the 5 thresholds ($t^* = 0.133$, yellow box) is applied to the test set.

old $t^* = 0.133$ it achieves sensitivity 100% with specificity 57.5% (a 38.5-point improvement in specificity over JSCCR at the same sensitivity level); at the default threshold (0.5) sensitivity drops to 56.2% but specificity rises to 92.5%.

PPV and prevalence. The PPV of 48.5% was computed with the test set’s prevalence (28.6% N1), higher than the real clinical rate (~ 8 –15%) [5]. At a prevalence of 10%, PPV drops to $\sim 21\%$ but NPV remains at 100%. The model is thus mainly useful as a **rule-out filter**: a negative result would allow surgery to be safely avoided. Sensitivity and specificity, independent of prevalence, are the more robust comparative metrics.

Comparison with other AI models. Piao et al.’s *gradient boosting* model [10] achieves better figures ($\text{spec.} = 85.8\%$, $\text{AUC} = 0.960$), but with $\sim 10\times$ more data and, crucially, from **clinicopathological variables already annotated by a pathologist**; our GAT uses only the WSI image and can run right after scanning. The fairer comparison, in the same modality (WSI without clinical variables), is Song et al. [27] (attention MIL over WSI, without clinical variables): over 1281 patients it achieves $\text{AUC} = 0.781$ and, at a practically identical specificity (57.6% vs 57.5%), $\text{sens.} = 92.9\%$, below our model’s 100%, although their evaluation is far more robust (1281 vs 56 patients). Representing the section as a spatial graph is thus at least competitive with flat MIL over *patches*.

TABLE 3: PERFORMANCE COMPARISON BETWEEN THE PROPOSED MODEL, CLINICAL GUIDELINES, AND THE STATE OF THE ART. EACH SYSTEM COMES FROM AN INDEPENDENT STUDY AND POPULATION (TANINO $n = 560$; IBC SCORE $n = 565$; LIGHTGBM $n = 105$; SONG $n = 1281$; GAT $n = 56$), SO PPV IS NOT DIRECTLY COMPARABLE. BAL.ACC. = $(S + E)/2$.

System	Input type	Sensitivity	Specificity	AUC	Bal.Acc.
<i>Guidelines and clinical scores</i>					
JSCCR [3]	Clinical guideline	100%	19%	0.596	59.5%
NCCN	Clinical guideline	98%	52%	0.749	75.0%
ESMO [9]	Clinical guideline	98%	50%	0.743	74.0%
IBC Score [4]	Histopath. score	90%	26%	0.68	58.0%
<i>AI models</i>					
LightGBM [10]	Clinicopathol. var.	100%	85.8%	0.960	92.9%
Song et al. [27]	WSI image	92.9%	57.6%	0.781	75.3%
<i>GAT Baseline (AUC = 0.875)</i>					
T1 ($t = 0.5$)	WSI image	56.2%	92.5%	0.875	74.4%
T2 ($t^* = 0.133$, TPR = 1)[†]	WSI image	100%	57.5%	0.875	78.8%
<i>GAT+DiffPool (AUC = 0.786)</i>					
T1 ($t = 0.5$)	WSI image	50.0%	82.5%	0.786	66.3%
T2 ($t^* = 0.303$) [†]	WSI image	62.5%	70.0%	0.786	66.3%

[†] t^* is the median of the maximum TPR = 1 thresholds across the 5 val sets, applied to the test set without readjustment.

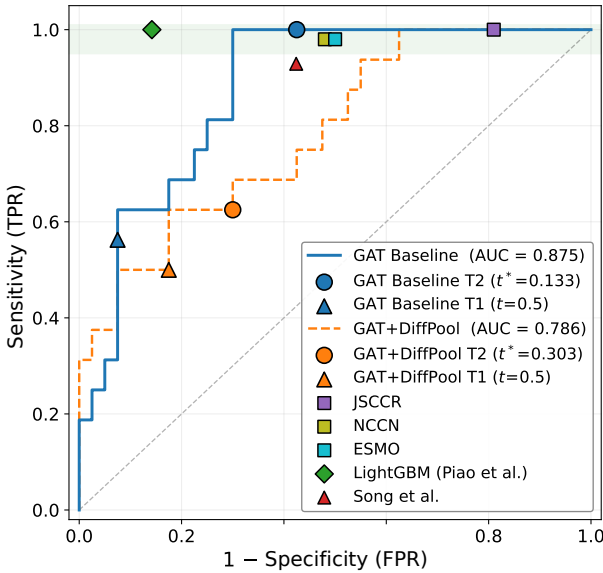


Fig. 6: ROC curves on the test set ($n = 56$) for the GAT Baseline (selected model; AUC = 0.875) and GAT+DiffPool (best configuration; AUC = 0.786). Circles mark each model’s t^* thresholds (derived from the val set). The operating points of JSCCR, NCCN, and ESMO (squares), *LightGBM* [10] (diamond), and Song et al. [27] (triangle) are overlaid.

8.2 Limitations

(1) **Test set size.** With 56 patients in the test set (only 16 N1), the 95% confidence intervals of the AUC are wide, with a margin of roughly ± 0.07 to ± 0.10 on each side of the estimate. The selected model’s CV AUC (0.862 ± 0.038) and test AUC (0.875) are consistent, but the variance of specificity across CV folds is high (mean 0.369 ± 0.198 at threshold t^*), reflecting instability across folds. A larger test set would be needed for robust clinical conclusions.

(2) **No external validation.** The dataset comes from 26

Spanish hospitals, but has not been tested on data from an independent center. Campanella et al. [28] report, in pathology MIL models, AUC drops of 3–6 points on external scanners and up to 7 points under institutional domain shift. Clinical applicability therefore requires external validation.

(3) **Prevalence imbalance.** The test set’s N1 prevalence (28.6%) exceeds the real clinical rate (~ 8 –15%); PPV must be interpreted with caution (see Discussion).

(4) **Threshold selection and val-set bias.** Two criteria were compared: fixed threshold $t = 0.5$ (T1) and the maximum threshold with TPR = 1 on val (T2, $t^* = 0.133$). T1 leaves FN = 7; T2 guarantees sens. = 100%. The val set had already been used for *early stopping*, so t^* may carry a slightly optimistic bias; ideally, it should be set on an external val set. The threshold’s variance across folds is high ($\sigma_{t^*} \approx 0.08$, folds: 0.03–0.29), reflecting the procedure’s sensitivity to the data split.

(5) **Comparison with methods evaluated on other datasets.** The clinical guidelines and AI models in Table 3 were evaluated on cohorts and populations different from ours, with disparate N1 prevalences and sample sizes, and were not applied to our test set. The comparison is therefore indicative rather than an apples-to-apples evaluation: sensitivity and specificity, independent of prevalence, are the most robust axes for comparison, but absolute differences must be interpreted with caution.

8.3 Conclusions

This thesis has built a complete N0/N1 classification system for pT1 colorectal cancer from WSI: generation of spatial Delaunay graphs, GAT model optimization with Bayesian TPE search, and final evaluation on a fixed test set (56 patients). The central question “**GAT Baseline or GAT+DiffPool?**” is answered clearly: **the GAT Baseline** (*attention pooling*, mean MIL, $h = 256$, $H = 4$, dropout 0.3) achieves the best balance: test AUC = 0.875, sensitivity 100% (all 16 N1 cases detected), and specificity 57.5%

at threshold $t^* = 0.133$ derived from the val set. Three observations stand out:

- **GAT Baseline outperforms GAT+DiffPool** on this dataset. Sections contain only a few hundred *patches*; hierarchical reduction to super-nodes discards more information than it synthesizes, and the best DiffPool stays at 0.786 test AUC, well behind the Baseline.
- **Prediction from the image alone.** The model diagnoses N0/N1 exclusively from the WSI, without any annotated clinicopathological variable (unlike models such as Piao et al.'s); processing is automatic from the scanned WSI (*patch* extraction, CLS embeddings, and graph construction), requiring neither the pathology report nor a manual pathologist score.
- **Usefulness as a rule-out filter.** At threshold t^* the model maintains sens. = 100% on the test set (NPV = 1.000): a negative result would allow surgery to be avoided with some confidence on the evaluated sample.

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APPENDIX

A.1 Example WSI Image and Segmentation Mask

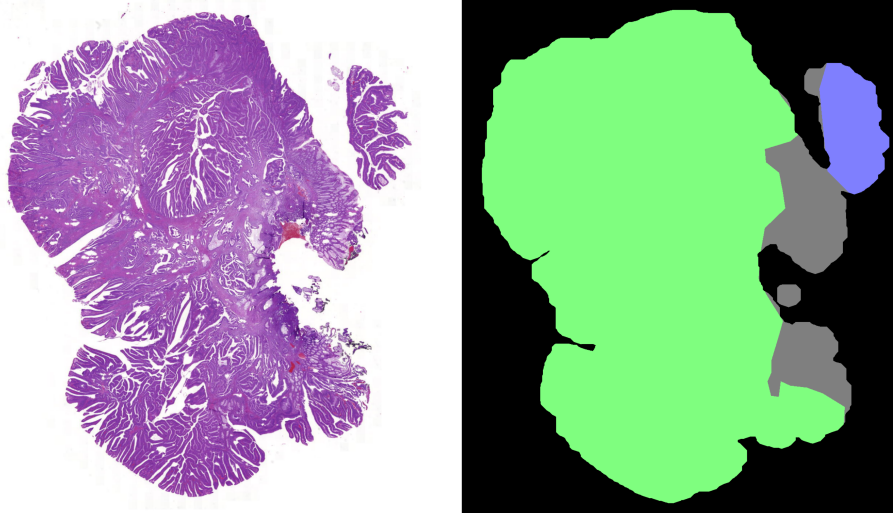


Fig. 7: Example WSI and its segmentation mask, with the annotated regions color-coded (see Section 4.1).

A.2 Detailed Pipeline Flow Diagram

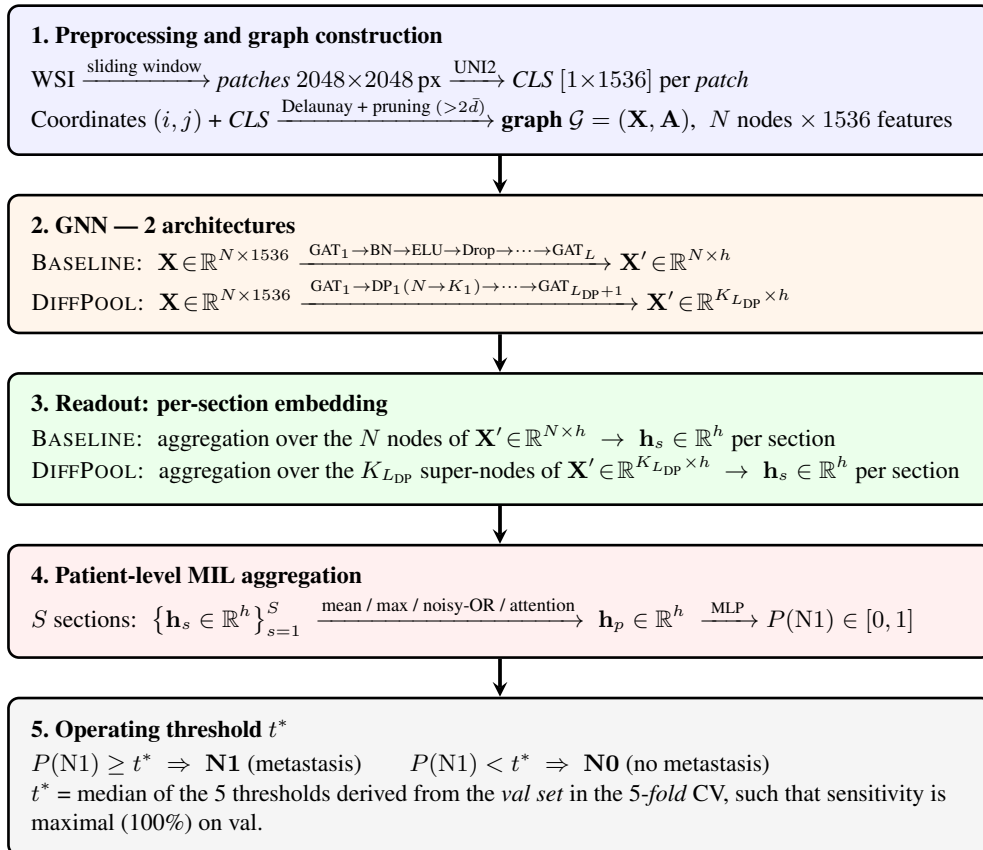


Fig. 8: Flow diagram of the complete *pipeline*, from the WSI to the N0/N1 decision. The five blocks are described in Section 5.1.

A.3 Architecture Variants of the *GATClassifier*

Diagram 9 shows how the **same input graph** can be processed in very different ways depending on the configuration: blue boxes represent learned layers and dashed orange boxes the configurable optimization axes.

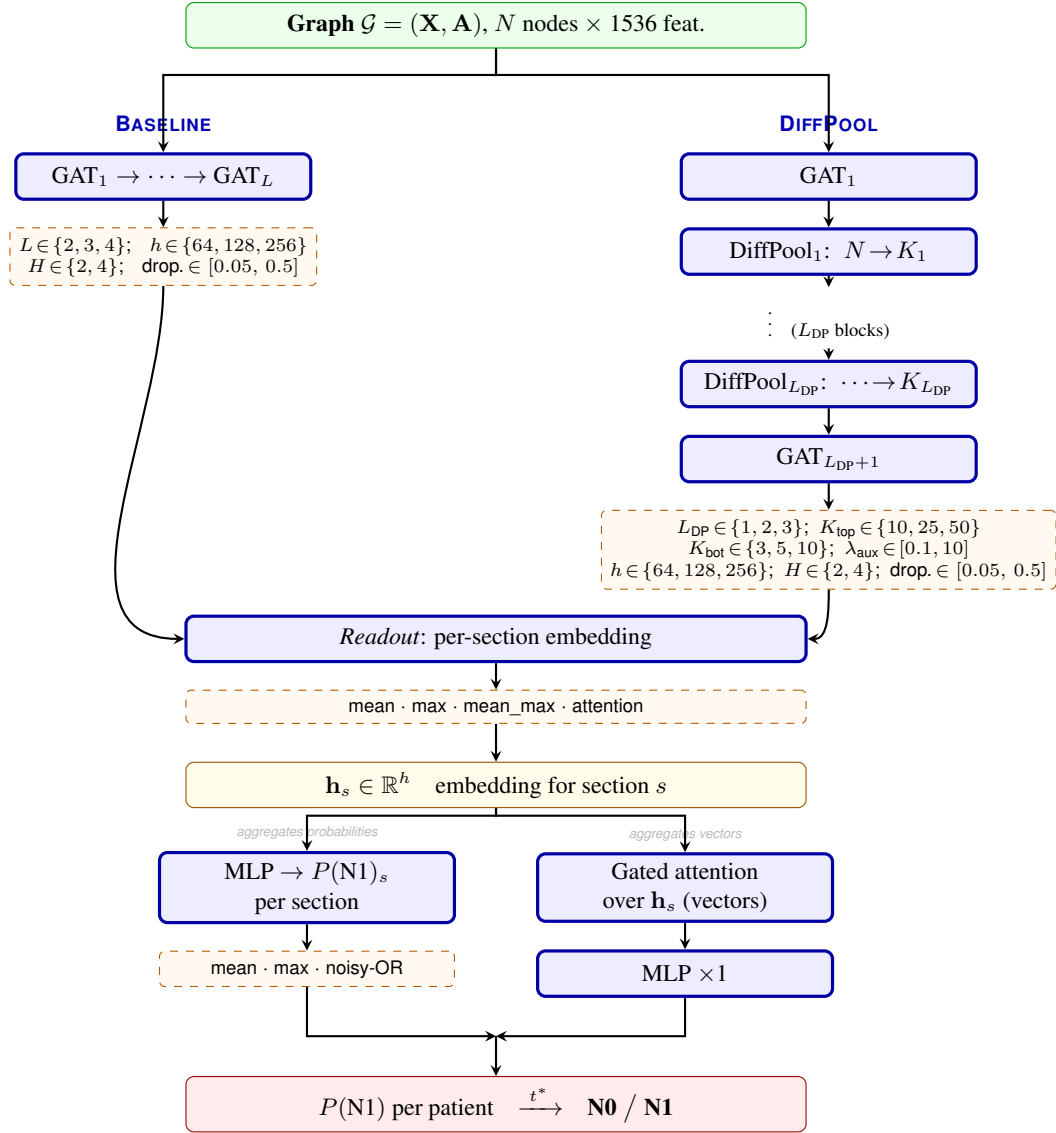


Fig. 9: Architecture variants of the *GATClassifier*: the same graph $\mathcal{G}$ is processed by two branches (BASELINE or DIFFPOOL) that converge into a unified *readout* ($\mathbf{h}_s \in \mathbb{R}^h$ per section) and then into MIL aggregation (Section 5.5). Symbols: L GAT layers; h hidden dim.; H attention heads; L_{DP} DiffPool blocks; K_ℓ super-nodes per level; λ_{aux} auxiliary loss weight; t^* val-set threshold.

A.4 Hyperparameter Importance per Architecture

Table 4 shows the **fANOVA importance** of each optimization axis, computed independently for each Optuna study (Baseline and DiffPool).

What is analyzed. The analyzed variable is each *trial*'s cross-validation objective ($\overline{\text{spec}}_{\text{CV}} - 10 \cdot \max(0, 1 - \overline{\text{sens}}_{\text{CV}})$, what the *sweep* maximizes). The *functional ANOVA* fits a surrogate model (*random forest*) over all *trials* and decomposes this objective's variance into each axis's contribution. The reported importance is the fraction of variance explained by the axis alone (main effect, in $[0, 1]$): the higher it is, the more that hyperparameter weighs on performance. They do not sum to 1 because interactions are not included. Axes specific to one branch appear empty in the other column.

TABLE 4: fANOVA IMPORTANCE PER ARCHITECTURE. EMPTY CELLS INDICATE THE AXIS DOES NOT APPLY TO THAT BRANCH.

Axis	Baseline	DiffPool
<i>Architecture-specific axes</i>		
L (GAT layers)	0.017	—
L_{DP} (DiffPool blocks)	—	0.062
K_{top}	—	0.027
K_{bot}	—	0.021
λ_{aux}	—	0.141
<i>Shared axes (optimized independently per study)</i>		
<i>Readout</i> (graph read-out)	0.079	0.053
h (hidden dim.)	0.012	0.027
H (attention heads)	0.024	0.066
<i>Dropout</i>	0.054	0.079
MIL aggregation	0.640	0.224
η (<i>learning rate</i>)	0.036	0.129
λ_{wd} (<i>weight decay</i>)	0.020	0.083
w_+ (<i>pos_weight</i>)	0.075	0.046
<i>Optimizer</i>	0.005	0.015
<i>Scheduler</i>	0.019	0.008
<i>Grad clipping</i>	0.017	0.019

A.5 Hyperparameters of the Selected Model and Best DiffPool

Table 5 summarizes the configuration of the GAT Baseline and GAT+DiffPool, already evaluated per the procedure in Table 2.

TABLE 5: CONFIGURATION OF THE GAT BASELINE AND GAT+DIFFPOOL.

Axis	Baseline	DiffPool
<i>Architecture — Baseline</i>		
L (GAT layers)	3	—
h / H / <i>Dropout</i>	256 / 4 / 0.3	—
<i>Readout</i>	<i>attention</i>	—
<i>Architecture — DiffPool</i>		
L ($= L_{\text{DP}} + 1$)	—	3
L_{DP} / K_{top} , K_{bot}	—	2 / 50, 3
h / H / <i>Dropout</i>	—	128 / 2 / 0.437
λ_{aux}	—	0.108
<i>Readout</i>	—	<i>mean_max</i>
<i>Training and regularization</i>		
MIL	<i>mean</i>	<i>mean</i>
η / λ_{wd} / w_+	$3 \cdot 10^{-4}$ / 10^{-4} / 2.0	$5.3 \cdot 10^{-4}$ / $5.5 \cdot 10^{-3}$ / 3.52
<i>Optimizer</i> / <i>Scheduler</i>	<i>adam</i> / <i>plateau</i>	<i>adam</i> / <i>cosine</i>
<i>Metrics re-evaluated on the test set</i>		
AUC CV ($\pm$ std)	0.862 ± 0.038	0.847 ± 0.048
AUC test	0.875	0.786
Sens. / Spec. test @ t^*	1.000 / 0.575	0.625 / 0.700
t^*	0.133	0.303

A.6 Bayesian Search Space

Table 6 lists the optimization axes, their ranges, and the architecture branch to which they apply. The search is done with *TPE* (Optuna); each *trial* evaluates a configuration with 5-fold CV over the train+val *pool*. The two architectures are optimized in **isolated Optuna studies** (separate SQLite databases): axes specific to each architecture are not mixed, and shared axes are optimized independently for each branch.

TABLE 6: BAYESIAN *sweep* AXES (TWO ISOLATED OPTUNA STUDIES, ONE PER ARCHITECTURE).

Axis	Values / range	Sampling type
<i>Model architecture — Baseline</i>		
L (GAT layers)	2, 3, 4	integer
<i>Readout</i>	mean, max, mean_max, attention	categorical
h (hidden dimension)	64, 128, 256	categorical
H (attention heads)	2, 4	categorical
<i>Dropout</i>	[0.05, 0.5]	uniform
<i>Model architecture — DiffPool</i>		
L_{DP} (DiffPool blocks)	1, 2, 3	integer
K_{top} (1st-level super-nodes)	10, 25, 50	categorical
K_{bot} (last-level super-nodes)	3, 5, 10	categorical
(Intermediate: $K = \lfloor \sqrt{K_{top} \cdot K_{bot}} \rfloor$; clamp $K_{bot} < K_{top}$)		
λ_{aux} (aux. weight)	[0.1, 10]	log-uniform
<i>Readout</i>	mean, max, mean_max, attention	categorical
h (hidden dimension)	64, 128, 256	categorical
H (attention heads)	2, 4	categorical
<i>Dropout</i>	[0.05, 0.5]	uniform
<i>MIL aggregation — both architectures (optimized independently)</i>		
MIL aggregation	mean, max, noisy-OR, attention	categorical
<i>Optimizer and regularization — both (independently)</i>		
<i>Optimizer</i>	Adam, AdamW	categorical
<i>Scheduler</i>	ReduceLROnPlateau, CosineAnnealingLR	categorical
Learning rate	$[5 \cdot 10^{-5}, 5 \cdot 10^{-3}]$	log-uniform
<i>Weight decay</i>	$[10^{-5}, 10^{-2}]$	log-uniform
<i>Grad. clipping</i>	0, 1, 5	categorical
N1 class weight (w_+)	[1.5, 8.0]	uniform

Sweep objective: $\text{objective} = \overline{\text{spec}}_{CV} - 10 \cdot \max(0, 1 - \overline{\text{sens}}_{CV})$, where $\overline{\text{spec}}_{CV}$ and $\overline{\text{sens}}_{CV}$ are the means over the 5 *val sets*, computed with the threshold t^* derived from each *fold* (highest threshold with TPR= 1). This is the **only** metric the TPE sees to decide the next *trial*'s parameters. The $\times 10$ penalty makes the TPE prioritize 100% sensitivity; any loss of sensitivity must be offset by a much larger specificity gain (10 points for every point of sensitivity lost).

A.7 Confusion Matrix of the Selected Model

Table 7 shows the confusion matrix of the selected GAT Baseline on the test set ($n=56$, $t^*=0.133$): all 16 N1 cases are detected (FN= 0, NPV= 1.000), and of the 40 N0 cases, 23 are classified correctly (17 FP, PPV= 0.485).

TABLE 7: CONFUSION MATRIX OF THE SELECTED GAT BASELINE ON THE TEST SET ($n=56$, $t^*=0.133$).

		Actual diagnosis	
		N0	N1
Prediction	N0	23	0
	N1	17	16